

Problem statement

Credit risk intelligence: loan default analytics for lending decisioning

1. Business context

Lending institutions — banks, NBFCs, and fintech lenders — approve loans based on a mix of credit bureau data, income verification, and an internally assigned risk grade. Once a loan is approved, the institution carries that risk on its books until the loan is repaid or defaults.

Loan defaults are one of the largest sources of financial loss in consumer lending. Even a small, undetected mispricing of risk across a large loan portfolio compounds into significant losses over time. At the same time, overly conservative underwriting turns away creditworthy borrowers and leaves revenue on the table. The core tension every lending business manages is: approve and price loans accurately enough to keep default losses low, without rejecting too many low-risk applicants.

2. The problem

This project analyzes a portfolio of 31,522 loan applications to answer a specific, practical question:

Which borrower and loan characteristics are actually associated with default, and is the current risk-grading and pricing structure aligned with real-world outcomes?

Three sub-problems motivate the analysis:

1. Risk concentration — is the loan book's dollar exposure concentrated in segments that are also the highest-risk, or is risk mismatched with where the money actually sits?
2. Pricing accuracy — do riskier loan purposes and borrower profiles carry appropriately higher interest rates, or is the current pricing model failing to compensate for the risk it's taking on?
3. Underwriting signal quality — which borrower attributes (loan grade, income, home ownership, debt-to-income ratio, credit history length, prior default history) actually predict default, and which commonly-assumed risk factors turn out not to matter once tested?

3. Why this matters to the business

- Loss prevention: identifying which segments defaults concentrate in allows underwriting teams to tighten approval criteria precisely, rather than applying blanket restrictions that also reject good borrowers.
- Revenue optimization: if certain loan purposes or grades are systematically underpriced relative to their real default risk, adjusting interest rates recovers margin without changing approval volume.
- Resource allocation: a small, prioritized list of the highest-risk existing loans lets a risk team focus manual review effort where it has the most impact, instead of reviewing the portfolio at random.
- Evidence-based policy: several commonly assumed risk factors (e.g., credit history length) are tested statistically rather than assumed - policies built on a factor that doesn't actually predict risk waste underwriting effort and may unfairly restrict credit access.

4. Objectives

4. Clean and validate a real-world loan applicant dataset, documenting every data quality decision made.
5. Quantify default rate across borrower and loan segments (grade, home ownership, loan purpose, income bracket, DTI, credit history).

6. Identify which numeric and categorical variables correlate most strongly with default, using both descriptive statistics and formal significance testing.
7. Engineer a simple, transparent, explainable composite risk score and validate that it actually predicts outcomes before recommending its use.
8. Store the analysis in a relational PostgreSQL database and answer business questions using advanced SQL (joins, CTEs, window functions).
9. Build an interactive Power BI dashboard that lets a non-technical stakeholder explore these findings without needing to read code.
10. Translate findings into specific, actionable underwriting and pricing recommendations.

5. Scope

In scope

- Exploratory and statistical analysis of the existing loan applicant dataset
- Descriptive risk segmentation and a rule-based (non-ML) composite risk score
- SQL-based business question answering
- Power BI dashboard for stakeholder-facing exploration

Out of scope

- Predictive machine learning models - deliberately excluded, since the business questions here are answerable with descriptive analytics and don't require a black-box model; a transparent, rule-based risk score was validated instead and found to perform well
- Real-time or streaming data pipelines - the dataset is static and historical
- Regulatory or compliance modeling (e.g., fair lending disparate-impact testing), which would require additional protected-class data not present in this dataset

6. Stakeholders

Stakeholder	What they need from this analysis
Credit risk team	Segment-level default rates to inform underwriting policy
Pricing / product team	Evidence on whether interest rates are risk-adjusted
Portfolio management	Exposure concentration by risk tier
Collections team	A prioritized list of existing high-risk loans

7. Dataset

- Source: Credit Risk Dataset (Kaggle, publicly available)
- Size: 32,581 raw records, 12 columns, one row per loan application
- Grain: one loan application per row
- Target / outcome variable: loan_status (0 = no default, 1 = default)

8. Success criteria

This analysis is considered successful if it produces:

- At least one statistically validated, non-obvious finding (not just a restatement of “riskier grades default more”)
- A working, correct SQL and Power BI layer that a reviewer can independently verify against the source data

- A short list of specific, numbers-backed recommendations a lending team could plausibly act on
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